GEOMATICS
NTNU

Version 03.2.2018

TBA4245 GEODESY

Example exercise no. 5: Gaussian conformal projection. Corrections

Solution on page 2-7.

Point V and R are two communication masts.
Geographical coordinates (in WGS84) and the corresponding UTM coordinates (the ellipsoidal heights are set to zero in the calculations):

Point	*Latitude, B*	*Longitude, L*	*N*	*E*
V	63.22354175° N	9.25197600° E Gr.w.	7010520.843	512665.511
R	63.88022131° N	11.41522236° E Gr.w.	7085912.227	618617.683

Azimuth from V to R is known: $\alpha_{V\text{-}R} = 54.79334266°$

(a) Correction of directions

Calculate the bearing on the plane from point R to V, $\varphi_{R\text{-}V} = t_{R\text{-}V}$, by using the known UTM coordinates in the table.

Use the bearing you calculated, $\varphi_{R\text{-}V}$, as a starting point for the following tasks:
Draw a figure and show the relationships between the azimuth, meridian convergence, plane bearing and correction of directions.
Calculate the azimuth from point R to point V, $\alpha_{R\text{-}V}$, by adding the necessary corrections to the plane bearing ($t_{R\text{-}V} = \varphi_{R\text{-}V}$). Figure!

(b) Geodesic

Calculate the azimuth of the geodesic from R to V, $\alpha_{R\text{-}V}$, by using formulas, which are valid for calculations on the ellipsoid.
Describe, without calculations, how the given value of the azimuth $\alpha_{V\text{-}R}$ can be computed.

(c) Scale factor

Calculate the scale factor in the two points V and R.

(d) Corrections of distances

Calculate the length of the geodesic between V and R.
Calculate the correction of distance for the distance between V and R
Calculate the distance on the map by correcting the length of the geodesic.
Calculate the distance on the map between the two points by using the UTM coordinates.
Compare the two calculated distances on the projection plane.

Solution:

Principle radii of curvature: $R_M = M$: Radius of curvature in meridian. $R_N = N$: Radius of curvature in prime vertical:

Point	R_M	R_N
V	6386485.54	6395221.28
R	6387074.92	6395418.00

NB: Remember the "UTM traps" with false origin of 500000 of the east coordinate and the scale factor of 0.9996:
Coordinates to be used in the corrections: Gaussian conformal map projection coordinates:

$$x_1 = x_R = N / 0.9996 = 7088747.726$$
$$x_2 = x_V = N / 0.9996 = 7013326.173$$
$$y_1 = y_R = \left(E - 500000\right) / 0.9996 = 118665.149$$
$$y_2 = y_V = \left(E - 500000\right) / 0.9996 = 12670.579$$

The sign of the corrections:

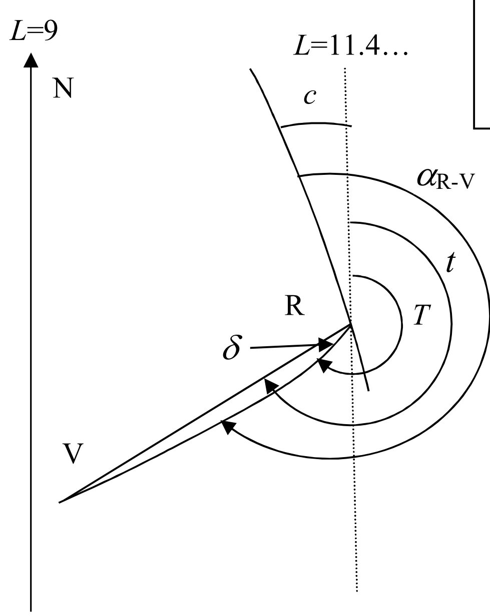
Normally, the sign is defined from correcting a value (azimuth or distance) <u>from</u> the ellipsoid <u>to</u> the plane.
But, check if this is the case before correcting!

(a)

Calculate the bearing on the plane from point R to V, $\varphi_{R\text{-}V} = t_{R\text{-}V}$, by using the known UTM coordinates in the table.

The tangent formula gives the answer (2 solutions, choose the right one!): $\tan(t_{\text{R-V}}) = {\Delta N_{\text{R-V}}}\big/{\Delta E_{\text{R-V}}}$

$\varphi_{\text{R-V}} = t_{\text{R-V}} = 260.62871827 \text{ gon} = 234.565846443°$



Draw a figure and show the relationships between the azimuth, meridian convergence, plane bearing and correction of directions.
Point R is in the UTM zone 32. See for instance the formula page in the prefix in Geodesy part 2. The Central meridian (N axis) is 9° East Greenwich Figure 4.3 in chapter 4 of Geodesy part 2.

Calculate the azimuth from point R to point V, $\alpha_{R\text{-}V}$, by adding the necessary corrections to the plane bearing ($t = \varphi_{R\text{-}V}$). Figure!

We must calculate c and δ:

(1) The meridian convergence, c, in point R: Here c is computed by using the formula given on the formula page in the prefix of "Geodesy part 2", where l = 11.41522236 - 9 = +2.41522236°:

$$c = l \cdot \sin B + (l^3/3) \cdot \sin B \cdot \cos^2 B \cdot (1 + 3 \cdot \varepsilon^2 + 2 \cdot \varepsilon^4) + (l^5/15) \cdot \sin B \cdot \cos^4 B \cdot (2 - \tan^2 B)$$

$$e^2 = (a^2 - b^2) / a^2 = 6.6943800047 \cdot 10^{-3}$$

$$\varepsilon^2 = \eta_R^2 = e^2 \cdot \cos^2 B_R \;\; / \;\; (1 - e^2) \;= e'^2 \cdot \cos^2 B_R = 1.30624461306 \cdot 10^{-3}$$

$$c = 2.1685693556° + 0.00024993 - 0.00000004 = 2.16881925° = +2.40979917 \text{ gon}$$

(which is Ok compared with the Holsen program KONVERG: 2.4097993 gon

From the figure, we see that the meridian convergence, c must be added to the calculated plane bearing t.

(2) The correction of horizontal directions, δ, **in point R:**
Correction from the ellipsoid to the plane, the Gaussian conformal projection.

The direction from R to V:
The radius of curvature in point R, the Gaussian mean, see "Geodesy part 1":

$$R_R = \sqrt{R_M \cdot R_N} \;= 6391245.10 \text{ m}$$

Formula (4.59), δ is defined as the bearing of the straight line R-V, t, minus the bearing of the image of the geodesic in point R, see also the figure above:

$$\delta_{R-V} = t_{R-V} - T_{R-V} = -\frac{\rho}{6R_R^2} \cdot (2y_R + y_V) \cdot (x_V - x_R) + \frac{\rho}{12R_R^4} \cdot (3y_R^2 \cdot y_V - y_R^3) \cdot (x_V - x_R)$$

$$\delta_{R-V} = +0.00489519 + 0.00000027 \;\; = +0.00489800 \text{ gon} = +0.00440820^{\circ}$$

⇨ From the figure we see that the correction of direction, δ, must be subtracted from the calculated plane bearing t. We go from the plane to the ellipsoid, the opposite way of how δ is defined.

By checking the signs of the corrections on the figure, we obtain:

The azimuth:

$\alpha_{\text{R-V}} = t_{\text{R-V}} + |c| - |\delta| =$
$= 260.62871827 + 2.40979917 - 0.00489800 = 263.0336194 \text{ gon} = 236.7302575°$

This numerical value agrees with the general formula without using the figure to set the signs of the corrections: $\delta = t - T$ gives $T = t - \delta$ and $\alpha = T + c$:

$\alpha_{\text{R-V}} = t_{\text{R-V}} + c - \delta =$
$= 260.62871827 + 2.40979917 - 0.00489800 = 263.0336194 \text{ gon} = 236.7302575°$

(b) *Calculate the azimuth of the geodesic from R to V, α_{R-V}, by using formulas, which are valid for calculations on the ellipsoid.*

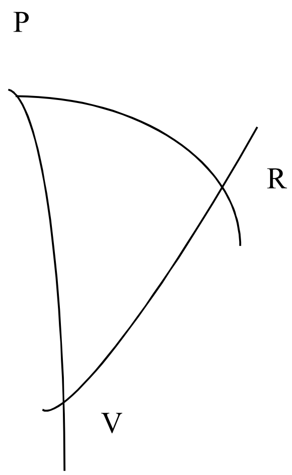
We know the latitudes of V and R and the azimuth $\alpha_{\text{V-R}}$ = 54.79334266°. Then it is possible to use one of Clairiaut's equations of the geodesic: $\cos\beta \cdot \sin\alpha = k$
Computing reduced latitudes:

$\beta = \tan^{-1}(\tan B \cdot (b / a))$, β_{V} = 63.14607082° and β_{R} = 63.80407978°

Calculating k in point V: $k = \cos \beta_V \cdot \sin\alpha_{V\text{-}R} = 0.369088376943$
Calculating the azimuth: $\alpha_R = \sin^{-1} (k / \cos\beta_R) = 56.7302569°$

Clairiaut's gives the azimuth from R and northwards, in the same direction as in point V.
We obtain the southward azimuth by correcting the calculated azimuth from Clairiaut's with 180°:
$\alpha_{R\text{-}V} = 236.73025690°$, which is the same value computed by using Holsen's program LGEOD-2 (The 2nd geodetic main task)

The answers in (a) and (b) give a difference of 0.000 00059 gon between the two calculated values of the azimuth. This is equal to a deviation (or an arc length) "tverravvik" of ca. 1,3 mm over a distance of 130 km. This is OK, is within the accuracy of the calculations and formulas.



Describe, without calculations, how the given value of the azimuth $\alpha_{V\text{-}R}$ can be computed.

Draw the triangle "the north pole P-point V-point R" on the ellipsoid. If we have minimum 3 known variables in this triangle, the other variables (sides and angles) can be computed. Here two sides are known, because the latitudes of V and R are known:
V-P = $(90 - B_V)$ and R-P = $(90 - B_R)$. Also the angle in P is known, the difference in longitudes $l = L_R - L_V$ The angle in point V can then be computed and this angle is the azimuth from V to R.

The solution is therefore to use the 2nd geodetic main task in chapter 2 in "Geodesy part 2":
Transform the given values on the ellipsoid to the Bessel sphere
Use spherical trigonometry
Transform the results from values on the sphere to values on the ellipsoid.

(c) Scale factor

Calculate the scale factor in the two points V and R.

Formula (4.58) in Geodesy part 2:

$$m = 1 + \frac{1}{2} \cdot \cos^2 B \cdot \left(1 + \eta^2\right) \cdot l^2 + \frac{1}{24} \cdot \cos^4 B \cdot \left(5 - 4t^2 + \cdots\right) \cdot l^4$$

$$\eta_V^2 = e'^2 \cdot \cos^2 B_V = 1.36784786627 \cdot 10^{-3}$$
$$\eta_R^2 = e'^2 \cdot \cos^2 B_R = 1.30624461306 \cdot 10^{-3}$$
$$e'^2 = \text{the 2nd numerical eccentrisity in square} = \frac{a^2 - b^2}{b^2}$$

Longitudes with the Central meridian (the north axis) as the zero meridian, UTM zone 32:
$l_V = L - 9 = +0.25197600°$
$l_R = L - 9 = +2.41522236°$

The scale factor in point V, the Gaussian projection:

$$m_V = 1 + 0.101618779817 \cdot l^2 - 1.837938 \cdot 10^{-2} \cdot l^4 =$$
$$= 1 + 1.9653824 \cdot 10^{-6} - 6.875 \cdot 10^{-12} =$$
$$= 1 + 0.0000019654 - 0.000000000 =$$
$$= 1.00000197$$

The scale factor in point R, the Gaussian projection:

$$m_R = 1 + 9.70362428532 \cdot 10^{-2} \cdot l^2 - 1.8215996 \cdot 10^{-2} \cdot l^4$$
$$= 1 + 1.72426078 \cdot 10^{-4} - 5.751629 \cdot 10^{-8} =$$
$$= 1 + 0.000172426 - 0.000000058 =$$
$$= 1.00017237$$

The scale factor in UTM:

$$m_{V,UTM} = m \cdot 0.9996 = 0.99960196$$
$$m_{R,UTM} = m \cdot 0.9996 = 0.99977230$$

Comments:
In UTM a scale factor between 0.9996 and 1.0004 is “allowed”. This means a maximum correction of 40 cm/km (40 ppm). However, the scale factor is larger than 1.0004 on the borderlines between UTM zones at Equator…, is Ok.
In Norway it is seldom that the scale factor is larger than 1.0000, because the zone becomes smaller as we go northwards, as all the meridians cross the North pole). *Exception: Between zones 33 and 35, as zone 34 is not used in Norway.*
The point R is ca. 2.4° from the north axis, ergo 0.6° from the borderline of the zones (6° broad zones in UTM), and has a scale factor which is less than 1.000000.
If we use l = 3° instead and the same latitude as R, the scale factor is:

$$m_{R,l=3,UTM} = (1 + 0.00026603 - 0.00000014) \cdot 0.9996 = 0.99986587 < 1$$

(d) Corrections of distances

Calculate the length of the geodesic between V and R.

$S_{GL} = 130081.072$ m (from the Geodetic main tasks)

Calculate the correction of distance for the distance between V and R
Calculate the distance on the map by correcting the length of the geodesic.

Correction of distance for the distance between V and R:

Formula (4.60a), $d = s - S$:

$$\boxed{d = \frac{s}{6 \cdot R_m^2} \cdot (y_1^2 + y_1 \cdot y_2 + y_1^2) - \frac{s \cdot \eta_m^2 \cdot \tan B_m}{6 \cdot R_m^3} \cdot (y_2^2 - y_1^2) \cdot (x_2 - x_1) - \frac{s \cdot 5 \cdot y_m^4}{24 \cdot R_m^4}}$$

Correction in the Gaussian conformal projection:

For corrected UTM coordinates, see page 2.

Radius of curvature, which radius shall we use when we are computing the Correction of distance?

Latitude of the middle point of the line:

$$B_m = \frac{1}{2}\cdot\left(B_V + B_R\right) = 63,55188153°$$

Gives the mean radius of curvature of the *mean point*, by first finding the principle radii of curvature, R_M and R_N at latitude B_m:

$$R_m = \sqrt{(R_M \cdot R_N)} = 6391049.351$$

$$\eta_m^2 = e'^2 \cdot \cos^2 B_m = 1.33691272491 \cdot 10^{-3}$$

$$e'^2 = \text{the 2nd numerical eccentrisity in square} = \frac{a^2 - b^2}{b^2}$$

$$t = \tan B$$

The distance in the formula is the distance on the plane, s. We could have used the ellipsoidal distance S as a very good approximate value, but for very long distances like here, we can also use an approximate value of s where only the first term is used (the sphere term): s = 130089.43 m

Entered into the correction formula (4.60a):

$$S_{R-V} - s_{R-V} = -8.3580 + 0.0002 = -8.3578$$

If we use ellipsoidal length, $S_G = 130081.072$ m, instead of the distance on the plane s:

$$S_{R-V} - s_{R-V} = -8.3575 + 0.0002 = -8.3573$$

Distance on the plane, Gaussian conformal projection (GCP):

$$s = S + 8.3578 = 130089.4298$$

2014: The last term in formula 4.60a:

$$\frac{s \cdot 5y_m^4}{24R_m^4} = 0,0003 \text{ m}$$

$$s = S + 8.3578 + 0,0003 = 130089.4301$$

Calculate the distance on the map between the two points by using the UTM coordinates.

We must calculate the UTM correction in addition to the GCP correction, see formula (4.60b):

$$\Delta s_{UTM} = s_{UTM} - s_{GCP} = -0.0004 \cdot s_{GCP} = -52.0358$$

Distance: $s_{UTM} = s - 52.0358 = 130037.3940$, or by 4.60b: 130037.3943

Compare the two calculated distances on the projection plane.

Calculate the distance on the plane from the UTM coordinates.

Pythagoras: $s = \sqrt{(\Delta N_{\text{R-V}})^2 + (\Delta E_{\text{R-V}})^2}$

$s = D_{kp} = 130037.3928$ OK, difference of 1,5 mm!